

Scenario 1: Image Classification Model Overfits Quickly:

You're training a CNN to classify dog breeds. You achieve 98% accuracy on training data within 5 epochs, but validation accuracy stays at 60%, and test performance is poor.

Logical Solution:

Problem: Overfitting due to complex model memorizing small dog breed training data.

Fixes:

1. Add Dropout(0.5) + BatchNorm after Conv/Dense layers
2. Data augmentation: RandomFlip, Rotation(15°), Brightness(±0.2)
3. EarlyStopping(patience=5) on val_loss + ReduceLROnPlateau
4. Transfer learning: VGG16/ResNet50 pretrained base (freeze early layers)

Scenario 2: ANN Fails to Learn Non-Linear Patterns

Project: Simple digit classification using **Artificial Neural Network (ANN)**

Problem: Accuracy stuck around 50%; adding more epochs doesn't improve performance.

Logical Solution:

Fixes:

1. Increase capacity: More neurons (128→512) + extra hidden layers (3→4)
2. Better activation: ReLU instead of sigmoid
3. Advanced optimizer: Adam(lr=0.001) → AdamW(weight_decay=1e-4) + LR scheduler
4. CNN instead: Replace ANN with Conv2D (digits need spatial patterns, not flatten)

Scenario 3: CNN Model Predicts Same Class for All Images

Project: Face Mask Detection using **CNN**

Problem: Model always predicts "With Mask" for every test image.

Logical Solution:

Fixes:

1. Class balance: Weighted loss ``class_weight={0:1.0, 1:3.0}`` or oversample minority
2. Gradient check: ``model.predict(random_images)`` → diverse outputs?
3. Focal loss or Label smoothing (0.1) → breaks prediction collapse
4. Monitor confusion matrix and recall/precision, not just accuracy.

Scenario 4: Object Detection Model Detects Objects but Bounding Boxes Are Misaligned

Project: YOLOv8 used for Tiger Detection in Forest

Problem: Bounding boxes do not fit objects properly.

Logical Solution:

Fixes:

1. Check labels: Make sure bbox outlines tigers exactly, not loose squares
2. Train longer: 100+ epochs for better box positioning
3. Add variety: Mix tiger sizes/angles/positions in training data
4. Adjust anchors: Use YOLO's auto-anchor feature for tiger shapes

Scenario 5: Face Mask Detection Gives High Accuracy but Performs Poorly on Real Faces

Project: Real-time Face Mask Detection using CNN

Problem: High accuracy on test set, but poor real-time performance on webcam feed

Logical Solution:

Fixes:

1. Real data: Add webcam-captured images to training set
2. Augmentation: Random brightness/contrast + blur + noise
3. Test preprocessing: Resize/crop webcam frames same as training (224x224)
4. Confidence threshold: Filter low-confidence predictions (0.7+)

Scenario 6: CNN-Based Object Detector Struggles at Night

Project: YOLOv8-based object detection in CCTV

Problem: Model doesn't detect objects in dark/night images.

Logical Solution:

Fixes:

1. Night data: Add dark CCTV images to training dataset
2. Augmentation: Random darkness (brightness 0.1-0.8) + noise
3. Image enhancement: CLAHE/Histogram equalization preprocessing
4. Infrared models: Train separate model on IR/thermal images